# SRINIVAS UNIVERSITY

# INSTITUTE OF ENGINEERING & TECHNOLOGY

**MUKKA, MANGALURU – 5741 46**

**MINIPROJECT REPORT - 1**

ON

# "Online Voting System using IBM Cloud Services"

***Submitted in the partial fulfillment of the requirements for the award of the degree of***

BACHELOR OF TECHNOLOGY IN
**CYBER SECURITY**

**Submitted By,**


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**2025-2026**

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## SRINIVAS UNIVERSITY, MUKKA

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# CERTIFICATE

This is to certify that the mini project entitled ***"Online Voting System using IBM Cloud Services"*** is a bonafide work carried out by ***Siddarth R, Skanda B,*** bearing the **USN** ***01SU23CB028, 01SU24CB701***in the partial fulfillment for the award of **Bachelor of Technology** in **Cyber Security** of the **Srinivas University, Institute of Engineering & Technology** during the year **2025-2026.** It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The mini project report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the said degree.

Mr. Yogesh Bandewar
IBM Trainer (Sri Thulasi EduTech Pvt.Ltd)
(Mini project Guide)

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# DECLARATION

We, **Siddarth R, Skanda B,** the student of sixth semester, **B.Tech** in Cyber Security, Srinivas University, Mukka, hereby declare that the mini project entitled ***"Online Voting System using IBM Cloud Services"*** has been successfully completed by us in partial fulfillment of the requirements for the award of degree in **Bachelor of Technology in Cyber Security of Srinivas University, Institute of Engineering and Technology** and no part of it has been submitted for the award of degree or diploma in any university or institution previously.

**Date:** _____

**Place:** Mukka

# ABSTRACT

This project presents the Online Voting System using IBM Cloud Services is a secure and scalable platform designed to conduct elections digitally. The system aims to eliminate the limitations of traditional voting methods by providing a convenient, transparent, and efficient way for users to cast their votes from anywhere.

This project leverages cloud-based technologies offered by IBM Cloud to ensure high availability, data security, and reliability. The system is built using a micro services architecture, where different components such as user authentication, voter management, candidate management, and vote processing operate independently. Services like cloud databases, identity authentication, and secure APIs are integrated to maintain data integrity and prevent unauthorized access.

The platform allows registered users to log in using secure credentials, view candidate details, and cast their vote only once. All votes are encrypted and stored securely in the cloud, ensuring privacy and preventing tampering. The system also provides real-time vote counting and result generation, improving efficiency and transparency.

Overall, the Online Voting System demonstrates how cloud computing can modernize electoral processes by making them more accessible, cost-effective, and secure.

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- **Siddarth R**
- **Skanda B**

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# Chapter 1

# INTRODUCTION

The rapid advancement of digital technologies has transformed various sectors, including banking, education, healthcare, and governance. One such area that can significantly benefit from digital transformation is the electoral process. Traditional voting systems, which rely on physical polling stations and manual vote counting, often face challenges such as long queues, human errors, lack of transparency, and high operational costs. To overcome these limitations, the concept of an Online Voting System has emerged as a modern solution.

An Online Voting System enables voters to cast their votes electronically through a secure internet-based platform. This system provides convenience by allowing users to participate in elections from any location, thereby increasing voter turnout and accessibility. However, ensuring security, privacy, and data integrity is crucial in such systems, as elections are highly sensitive processes.

This project focuses on developing an Online Voting System using IBM Cloud Services, which provides a reliable and scalable cloud infrastructure. By utilizing cloud technologies, the system ensures high availability, secure data storage, and efficient handling of large volumes of users. The system is designed using a microservices architecture, where different functionalities such as authentication, vote management, and result processing are handled independently.

The platform includes features such as user registration, secure login, candidate listing, vote casting, and real-time result generation. Advanced security mechanisms like encryption and authentication are implemented to prevent unauthorized access and ensure that each voter can vote only once. The use of IBM Cloud services enhances the system's reliability and scalability, making it suitable for large-scale deployment.

Overall, the Online Voting System aims to provide a secure, transparent, and efficient alternative to traditional voting methods, contributing to the modernization of electoral processes.

## 1.1 OBJECTIVES

- To develop a secure and reliable online platform for conducting elections digitally.
- To eliminate the limitations of traditional voting methods such as long queues, manual errors, and time consumption.
- To ensure that each voter can cast their vote only once using proper authentication mechanisms.
- To maintain the confidentiality and integrity of voting data through encryption and secure storage on the cloud.
- To provide an easy-to-use interface for voters to register, log in, and cast their votes from any location.
- To implement a scalable system using IBM Cloud Services that can handle a large number of users efficiently.
- To enable real-time vote counting and instant result generation.
- To provide an admin module for managing voters, candidates, and election processes.

- To ensure transparency and fairness in the election process.

## 1.2 Overview

The Online Voting System using IBM Cloud Services is a web-based application designed to simplify and secure the election process through digital technology. The system allows registered users to participate in voting from any location using an internet connection, thereby eliminating the need for physical polling stations.

The platform is developed using cloud infrastructure, which ensures high availability, scalability, and data security. IBM Cloud Services are used to host the application, manage databases, and provide secure authentication mechanisms. The system follows a microservices architecture, where different components such as user management, authentication, voting process, and result processing operate independently.

The system consists of two main users: voters and administrators. Voters can register, log in securely, view the list of candidates, and cast their vote. Once a vote is submitted, it is securely stored in the cloud and cannot be modified. The system ensures that each voter is allowed to vote only once.

Administrators are responsible for managing the overall system, including adding candidates, verifying voters, starting or ending the election process, and viewing results. The system also provides real-time vote counting and result generation, improving efficiency and transparency.

Overall, this project demonstrates how cloud computing and modern web technologies can be used to create a secure, efficient, and user-friendly online voting platform.

## 1.3 Problem Statement

Traditional voting systems face several challenges that affect the efficiency, security, and accessibility of the electoral process. These systems require physical presence at polling stations, which can be inconvenient for voters who are located far away, have mobility issues, or face time constraints. As a result, voter participation may be reduced.

Manual voting methods are also prone to errors during vote casting and counting, leading to inaccurate results. Additionally, the process of managing large-scale elections involves high costs, extensive manpower, and significant time consumption. Ensuring transparency and preventing fraudulent activities such as duplicate voting or vote tampering is another major concern in traditional systems.

Furthermore, maintaining and securing physical records is difficult and increases the risk of data loss or manipulation. In today’s digital era, there is a need for a more efficient, secure, and reliable system that can overcome these limitations.

Therefore, it is essential to develop an Online Voting System that leverages cloud technology to provide secure authentication, protect voter data, ensure one-person-one-vote, and enable fast and accurate result processing while improving accessibility and transparency.

## 1.4 Solution

To overcome the limitations of traditional voting systems, an Online Voting System using IBM Cloud Services is proposed as an effective solution. This system provides a secure, efficient, and user-friendly platform for conducting elections digitally.

The proposed system allows voters to register and log in using authenticated credentials. Once verified, users can access the voting portal, view the list of candidates, and cast their vote from any location. The system ensures that each voter can vote only once by implementing strict authentication and validation mechanisms.

IBM Cloud Services are utilized to host the application and manage data securely. Cloud-based storage ensures data integrity, availability, and protection against unauthorized access. Advanced security features such as encryption and secure APIs are used to safeguard sensitive voter information and prevent vote tampering.

The system follows a microservices architecture, where different modules such as user management, voting process, and result analysis operate independently. This improves scalability and makes the system capable of handling large numbers of users efficiently.

Additionally, the system provides real-time vote counting and instant result generation, reducing manual effort and increasing transparency. An admin panel is included to manage voters, candidates, and election activities effectively.

Overall, this solution offers a modern approach to elections by making the voting process more accessible, reliable, and secure while reducing cost and time.

# Chapter 2

## LITERATURE SURVEY

The concept of online voting systems has gained significant attention in recent years due to the increasing demand for secure and efficient digital solutions in electoral processes. Various researchers and organizations have explored different approaches to develop reliable online voting platforms.

Earlier voting systems were primarily paper-based, which involved manual vote casting and counting. These systems were time-consuming, error-prone, and required significant manpower. To overcome these issues, Electronic Voting Machines (EVMs) were introduced, which improved accuracy and reduced counting time. However, EVMs still required voters to be physically present at polling stations and raised concerns regarding security and transparency.

With the advancement of internet technologies, web-based voting systems were developed to allow remote voting. Several studies highlight the importance of authentication mechanisms such as username-password login, biometric verification, and two-factor authentication to ensure that only authorized users can vote. Security techniques like encryption and secure communication protocols are widely used to protect voter data and maintain confidentiality.

Recent research focuses on the use of cloud computing in online voting systems. Cloud platforms provide scalability, flexibility, and cost-effectiveness, making them suitable for large-scale elections. IBM Cloud Services, in particular, offer secure infrastructure, identity management, and database solutions that enhance system reliability and data protection.

Some modern approaches also explore blockchain technology to ensure transparency and immutability of votes. While blockchain-based systems provide strong security features, they may introduce complexity and higher implementation costs.

Overall, the literature suggests that combining cloud computing with strong security mechanisms can significantly improve the efficiency, accessibility, and reliability of online voting systems. The proposed system builds upon these concepts by utilizing IBM Cloud Services to develop a secure and scalable online voting platform.

# Chapter 3

# REQUIREMENT SPECIFICATIONS

Requirement specification defines the functional and non-functional requirements necessary for the development and operation of the Online Voting System. It outlines what the system should do and the constraints under which it must operate.

## Functional Requirements

The functional requirements describe the core functionalities of the system:

- The system must allow users to register with valid details such as name, email ID, and password.
- The system must provide secure login authentication for registered users.
- The system must allow the administrator to add, update, and delete candidate details.
- The system must display the list of candidates to eligible voters.
- The system must allow each voter to cast their vote only once.
- The system must store votes securely in the cloud database.
- The system must provide real-time vote counting and result generation.
- The system must allow the administrator to manage voters and monitor the election process.
- The system must generate reports of voting results.

## Non-Functional Requirements

The non-functional requirements define the quality attributes of the system:

- **Security:** The system must ensure data encryption, secure authentication, and protection against unauthorized access.
- **Performance:** The system should handle multiple users simultaneously without delay.
- **Scalability:** The system must be able to scale efficiently using IBM Cloud Services.
- **Reliability:** The system should operate continuously without failure during elections.
- **Usability:** The user interface should be simple and easy to use for all users.
- **Availability:** The system must be accessible at all times during the voting period.

## Hardware Requirements

- Computer/Laptop with minimum 4 GB RAM
- Internet connection
- Server (Cloud-based – IBM Cloud)

## Software Requirements

- Operating System: Windows/Linux
- Frontend: HTML, CSS, JavaScript
- Backend: Node.js / Java / Python (any suitable technology)
- Database: IBM Cloud Database (Cloudant / Db2)
- Tools: Visual Studio Code, Web Browser

# Chapter 4

# SYSTEM DESIGN

System design describes the overall architecture, components, and workflow of the Online Voting System. It provides a structured approach to developing the system using cloud-based technologies and microservices architecture.

## 4.1 Project Modules

The system is divided into several modules, each responsible for a specific functionality:

- **User Registration Module:** Allows new users to register by providing required details.
- **Authentication Module:** Handles secure login and verification of users.
- **Voter Module:** Enables voters to view candidates and cast their vote.
- **Admin Module:** Allows administrators to manage voters, candidates, and election processes.
- **Voting Module:** Manages vote casting and ensures that each user votes only once.
- **Result Module:** Handles vote counting and displays election results.

## 4.2 System Architecture

The system follows a **microservices architecture** deployed on IBM Cloud Services. Each service operates independently and communicates through secure APIs.

- **Frontend Layer:** User interface developed using web technologies for interaction with users.
- **API Gateway:** Acts as an entry point for all client requests and routes them to appropriate services.
- **Microservices Layer:** Includes separate services for authentication, voting, user management, and results.
- **Database Layer:** Cloud-based database (IBM Cloudant/Db2) used to store user data, votes, and results securely.
- **Security Layer:** Implements encryption, authentication, and access control mechanisms.

This architecture ensures scalability, flexibility, and fault tolerance.

## 4.3 System Workflow

The workflow of the system is as follows:

1. User registers on the platform with valid details.
2. User logs in using secure credentials.
3. The system verifies the user through the authentication service.
4. After successful login, the voter can view the list of candidates.
5. The voter selects a candidate and casts the vote.
6. The system validates that the user has not voted before.
7. The vote is securely stored in the cloud database.

# Chapter 5

# MODULES OF THE SYSTEM

The Online Voting System is divided into several modules to ensure smooth functioning and easy management of the entire application. Each module performs a specific task and works independently within the system.

**User Registration Module**

This module allows new users to register in the system by providing necessary details such as name, email ID, and password. The system validates the information and stores it securely in the cloud database.

**Login and Authentication Module**

This module is responsible for verifying user credentials. It ensures that only registered users can access the system. Secure authentication techniques are used to protect user accounts and prevent unauthorized access.

**Voter Module**

The voter module allows authenticated users to participate in the election process. Voters can view the list of candidates, their details, and cast their vote. The system ensures that each voter can vote only once.

**Admin Module**

The admin module provides control over the entire system. The administrator can add, update, or delete candidate details, manage voter records, start or end the election, and monitor the voting process.

**Voting Module**

This module handles the core functionality of vote casting. It records the votes securely and ensures that no duplicate voting is allowed. The module also ensures that the voting process is confidential and tamper-proof.

**Result Module**

The result module is responsible for counting votes and generating results. It provides real-time updates and displays the final outcome of the election in a clear and transparent manner.

# Chapter 6

## TESTING

Testing is an essential phase in the development of the Online Voting System to ensure that the application functions correctly, securely, and efficiently. It helps identify and fix errors before the system is deployed for real-time use.

### Unit Testing

Unit testing is performed on individual modules such as registration, login, voting, and result processing. Each module is tested separately to verify that it works as expected.

### Integration Testing

Integration testing ensures that all modules of the system work together properly. It checks the interaction between components such as the frontend, backend services, and cloud database.

### System Testing

System testing is conducted on the complete application to verify that the entire system meets the specified requirements. It ensures that all functionalities work correctly in a real-world environment.

### Security Testing

Security testing is crucial for an online voting system. It ensures that user data and votes are protected from unauthorized access, hacking, and tampering. Techniques such as authentication checks and data encryption are verified.

### Performance Testing

Performance testing evaluates how the system behaves under different loads. It ensures that the application can handle multiple users simultaneously without slowing down or crashing.

### User Acceptance Testing (UAT)

User Acceptance Testing is performed to ensure that the system meets user expectations and requirements. Real users test the system and provide feedback on usability and functionality.

# Chapter 7

## IMPLEMENTATION

Implementation is the phase where the designed system is developed and deployed using appropriate technologies and tools. In this project, the Online Voting System is implemented using web technologies and IBM Cloud Services to ensure scalability, security, and reliability.

The system is developed using a client-server architecture. The frontend is designed using HTML, CSS, and JavaScript to provide a user-friendly interface for voters and administrators. The backend is implemented using technologies such as Node.js / Java / Python, which handle business logic, authentication, and communication with the database.

IBM Cloud Services are used to host the application and manage the database. Cloud-based databases such as IBM Cloudant or Db2 are used to store user information, candidate details, and voting records securely. The use of cloud infrastructure ensures high availability and efficient data management.

The application is divided into multiple microservices such as user management, authentication, voting, and result processing. These services communicate through secure APIs, making the system modular and easy to maintain.

During implementation, security measures such as password encryption, secure login authentication, and access control are integrated to protect sensitive data. The system also includes validation mechanisms to ensure that each voter can vote only once.

The admin panel is implemented to allow administrators to manage users, candidates, and election activities. Features such as adding candidates, monitoring votes, and generating results are included.

Finally, the system is deployed on IBM Cloud, making it accessible through the internet. Proper testing is conducted before deployment to ensure that the system works efficiently without errors.

# Chapter 8

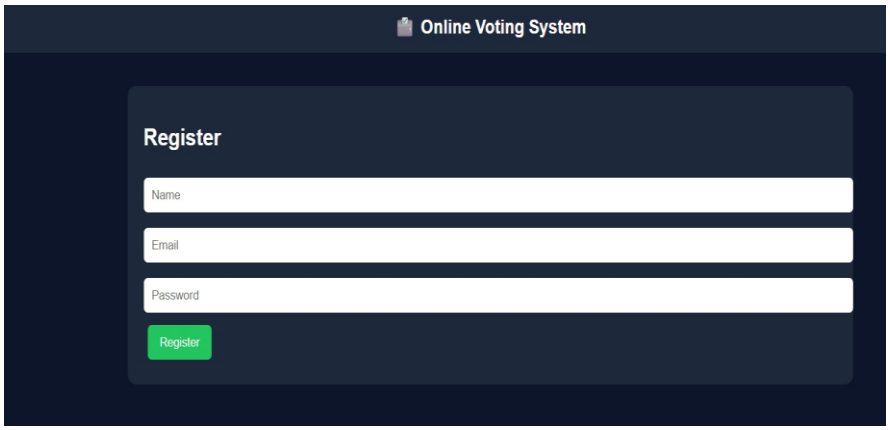
## RESULTS

The Online Voting System developed using IBM Cloud Services was successfully implemented and tested. The system provides a secure, efficient, and user-friendly platform for conducting elections digitally. All modules were integrated and tested to ensure smooth functionality and accurate performance.

### 8.1 System Overview Interface

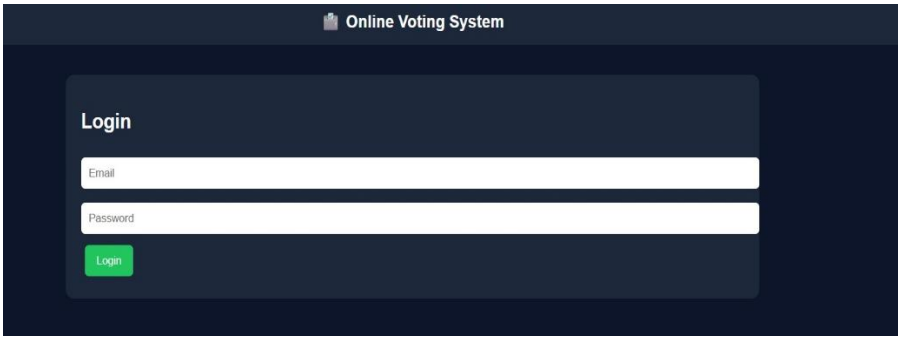
The system provides a simple and intuitive user interface for both voters and administrators. Users can easily navigate through registration, login, and voting pages without difficulty.

### 8.2 User Registration



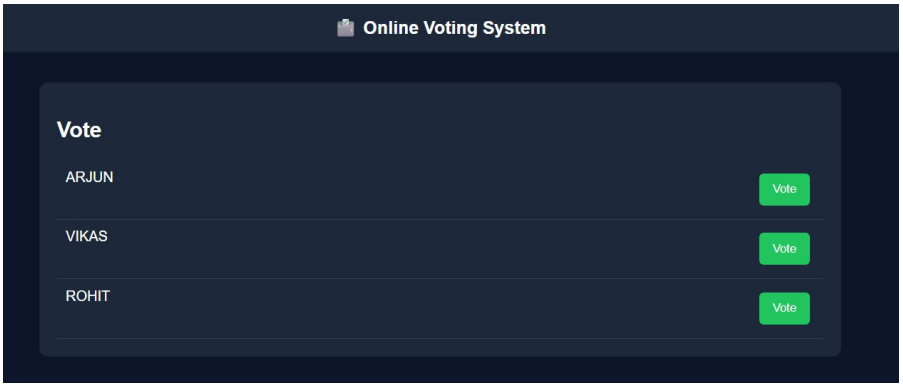

New users can register by entering valid details such as name, email ID, and password. The system validates the information and stores it securely in the cloud database.

### 8.3 Login Authentication




Registered users can log in using their credentials. The system verifies user identity and allows access only to authorized users, ensuring security.

### 8.4 Voting Interface

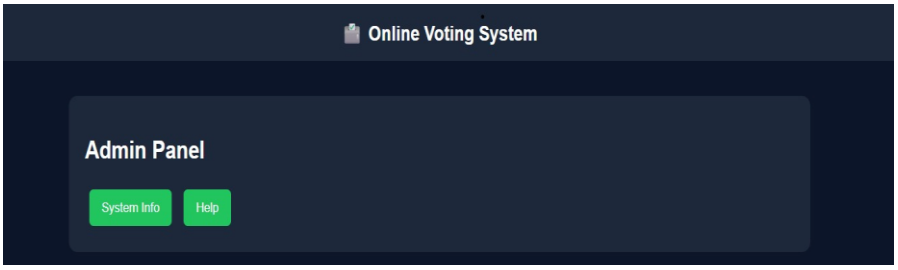



After successful login, voters can view the list of candidates along with their details. The interface allows users to select their preferred candidate and cast their vote easily.

### 8.5 Vote Confirmation

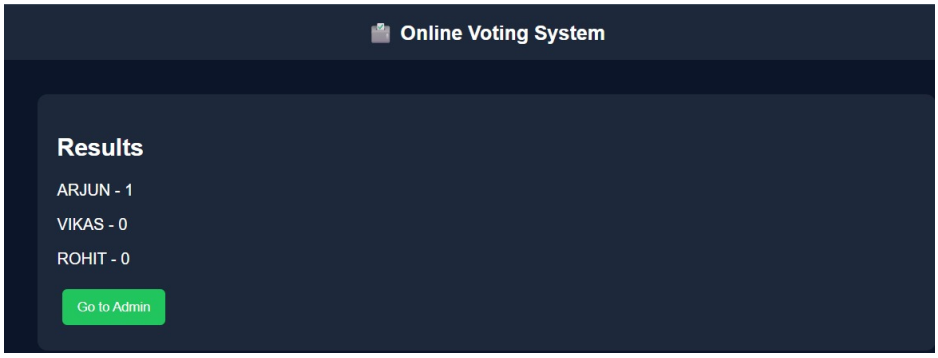
Once a vote is cast, the system confirms the submission and ensures that the user cannot vote again. This maintains the integrity of the election process.

### 8.6 Admin Panel




The admin dashboard allows administrators to manage candidates, monitor voting activities, and control the election process. It provides real-time updates on voting progress.

## 8.7 Result Display




The system generates results instantly after voting ends. Vote counts are displayed clearly, ensuring transparency and accuracy.

# Chapter 9

## CHALLENGES AND LIMITATIONS

The development and implementation of the Online Voting System involve several challenges and limitations that need to be considered to ensure system effectiveness and reliability.

### Challenges

- **Security Concerns:** Ensuring the protection of sensitive voter data and preventing unauthorized access or cyber-attacks is a major challenge.
- **Authentication Issues:** Verifying the identity of users accurately to prevent duplicate or fake voting can be difficult.
- **Internet Dependency:** The system requires a stable internet connection, which may not be available in all areas.
- **Scalability Challenges:** Handling a large number of users simultaneously during peak voting time can affect system performance.
- **System Reliability:** Ensuring continuous system availability without downtime during elections is critical.
- **User Awareness:** Some users may not be familiar with digital platforms, making it difficult for them to use the system effectively.

### Limitations

- The system depends completely on internet connectivity and cloud services.
- It may be vulnerable to technical failures such as server downtime or network issues.
- Security risks such as hacking, phishing, or data breaches cannot be completely eliminated.
- Not suitable for areas with low digital literacy or limited access to technology.
- Requires proper maintenance and regular updates to ensure system performance and security.

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# Chapter 10

## FUTURE ENHANCEMENTS

The Online Voting System can be further improved by incorporating advanced features and technologies to enhance security, usability, and performance.

- **Biometric Authentication:** Implement fingerprint or facial recognition to strengthen user verification and prevent unauthorized access.
- **Two-Factor Authentication (2FA):** Add an extra layer of security using OTP-based verification during login.
- **Blockchain Integration:** Use blockchain technology to ensure transparency, immutability, and tamper-proof voting records.
- **Mobile Application Support:** Develop a mobile app to make the system more accessible and user-friendly.
- **Multi-language Support:** Provide support for multiple languages to make the system usable for a wider audience.
- **Offline Voting Support:** Introduce mechanisms to allow voting in low-connectivity areas with later synchronization to the cloud.
- **Advanced Analytics Dashboard:** Enhance the admin panel with detailed analytics and graphical reports.
- **AI-based Fraud Detection:** Implement artificial intelligence techniques to detect suspicious activities and prevent fraud.
- **Scalability Improvements:** Optimize the system to handle national-level elections with a very large number of users.

# Chapter 11

## CONCLUSION

The Online Voting System using IBM Cloud Services successfully demonstrates how modern technology can be utilized to improve the traditional voting process. The system provides a secure, efficient, and user-friendly platform that allows voters to participate in elections from any location.

By leveraging cloud computing, the system ensures high availability, scalability, and reliable data storage. The implementation of authentication mechanisms and security measures helps in maintaining the confidentiality and integrity of voter information and voting data. The system also ensures that each voter can cast their vote only once, thereby maintaining fairness in the election process.

The use of a microservices architecture makes the system modular, flexible, and easy to maintain. Features such as real-time vote counting, instant result generation, and an admin dashboard enhance transparency and efficiency.

Although there are certain challenges and limitations, the system provides a strong foundation for conducting digital elections. With further improvements and advanced technologies, it has the potential to be used in large-scale and real-world election scenarios.

In conclusion, the Online Voting System is a significant step towards digital transformation in the electoral process, making voting more accessible, reliable, and secure.

# Chapter 12

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